

Table 1: GCI–Richardson extrapolation vs. Abaqus ($n = 500$): deflection and rotation.

Job	Load type	QoI	n_{fine}	p_{obs}	ϕ_{fine}	ϕ_{ext}	Abaqus ($n = 500$)	Error ext. (%)	GCI ₁₂ (%)	GCI ₂₃ (%)
0	End	u_y (mm)	128	-3.58	-3.0413	-3.0413	-3.0342	0.234	—	—
0	End	θ_z (deg)	128	-3.97	-0.1307	-0.1307	-0.1302	0.401	—	—
1	Mid-span	u_y (mm)	128	-2.61	-0.9504	-0.9504	-0.9468	0.376	—	—
1	Mid-span	θ_z (deg)	128	-3.42	-0.0327	-0.0327	-0.0324	0.805	—	—
2	Quarter-span	u_y (mm)	128	—	-0.2614	-0.2614	-0.2596	0.686	—	—
2	Quarter-span	θ_z (deg)	128	—	-0.0082	-0.0082	-0.0080	1.619	—	—
5	UDL	u_y (mm)	128	2.00	-1.3179	-1.3179	-1.3143	0.270	0.0015	0.0059
5	UDL	θ_z (deg)	128	2.00	-0.0523	-0.0523	-0.0520	0.502	0.0030	0.0119
6	Triangular	u_y (mm)	128	2.00	-3.0525	-3.0526	-3.0342	0.607	0.0019	0.0076
6	Triangular	θ_z (deg)	128	0.88	-0.1307	-0.1307	-0.1302	0.401	0.0000	0.0000
7	Parabolic	u_y (mm)	128	2.00	-0.9560	-0.9560	-0.9468	0.973	0.0030	0.0121
7	Parabolic	θ_z (deg)	128	0.74	-0.0327	-0.0327	-0.0324	0.805	0.0000	0.0000

Note. FEM grids: $n_1 = 128$, $n_2 = 64$, $n_3 = 32$ elements (refinement ratio $r = 2$); safety factor $F_s = 1.25$.

Abaqus reference: 500-element mesh (converged benchmark). ϕ_{fine} = fine-grid FEM solution; ϕ_{ext} = Richardson extrapolate. Error ext. = $100(\phi_{\text{ext}} - \text{Abaqus})/|\text{Abaqus}|$. GCI₁₂, GCI₂₃ from three-grid formula. Tip deflection in mm, tip rotation in deg. Abaqus tip rotation is zero if UR was not in the ODB.